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ORIGINAL ARTICLE

Subgingival microbiota in individuals with severe chronic periodontitis

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Abstract *Background/Purpose:* Subgingival microorganisms are potentially associated with periodontal diseases. However, the correlation between the variance in the periodontal microbiome and the prevalence and severity of periodontitis remains unclear. The aim of this study was to determine the subgingival microbiota in Taiwanese individuals with severe chronic periodontitis (SP).

Methods: The composition of the subgingival microbiota in healthy and diseased individuals was compared using a 16S rRNA metagenomic approach and quantitative polymerase chain reaction (qPCR). A total of 20 samples, including 10 from healthy individuals and 10 from SP patients, were analyzed.

Results: We found high microbial diversity, with an average of 774 classified phylotypes per sample and a total of six bacterial phyla across all samples. Cluster analysis by principal component analysis and heat map showed that the bacterial communities were different in the two groups. *Streptococcus* dominated across all the healthy samples, whereas *Prevotella*, *Porphyromonas*, and *Treponema* were highly abundant across all diseased samples. At least 13 bacterial genera were conserved among all the samples. Only eight genera, including *Lautropia*, *Parvimonas*, *Actinomyces*, *Capnocytophaga*, *Paludibacter*, *Streptococcus*, *Haemophilus*, and *Corynebacterium*, were significantly enriched in the healthy group, and six genera, including *Porphyromonas*, *Treponema*, *Tannerella*, *Aggregatibacter*, *Peptostreptococcus*, and *Filifactor*, were significantly enriched in the diseased group. Furthermore, a trend of abundance of bacteria at the species level measured by qPCR in all samples was consistent with the 16S rRNA metagenomics results.

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Conclusion: This study is the first in Taiwan to provide a picture of the microbiome in SP via 16S rRNA metagenomics.
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Introduction

Chronic periodontitis is defined as destruction of the periodontal ligament and loss of the adjacent supporting bone, which may be due to inflammation of the gingiva and the adjacent attachment apparatus.[1] Periodontitis can be further classified on the basis of severity and extent. For severity, periodontitis is classified into three levels—slight, moderate, and severe destruction—according to the loss of periodontal attachment.[1] Severe chronic periodontitis (SP) is characterized as clinical attachment loss (CAL) of $\geq$5 mm. Left untreated, chronic periodontitis, especially SP, can lead to or aggravate existing conditions such as diabetes, cardiovascular disease, pulmonary diseases, and obesity.[2,3]

Periodontal diseases are very common, affecting up to 90% of the global population. The common inflammatory forms of the diseases are caused by pathogenic microflora in the biofilm or dental plaque that forms adjacent to the teeth on a daily basis and can result in loosening of teeth, occasional pain, and eventual tooth loss. Several Gram-negative bacteria, including *Porphyromonas gingivalis*, *Treponema denticola*, and *Tannerella forsythia*, are frequently isolated from dental plaques in periodontal patients and were initially considered periodontal pathogens.[4] A strong correlation between several cultivable bacteria such as *Prevotella intermedia*, *Fusobacterium nucleatum*, *Aggregatibacter actinomycetemcomitans*, and *Eubacterium nodatum* and periodontal disease has also been reported.[5,6] Although subgingival bacteria are the major cause of periodontal diseases,[4] more than one-half of subgingival bacterial species or phylotypes are not readily cultivable,[7] which presents an obstacle to fully understand the causal relationships between subgingival bacteria and periodontitis. To overcome the difficulties and limitations associated with cultivation, culture-independent methods based on amplification and sequencing of bacterial metagenomes have been developed to identify thousands of different bacteria in a single sample.[8]

Recently, several reports focused on subgingival bacterial diversity using 16S rRNA amplicon metagenomics or whole genome shotgun sequencing (WGS).[9–12] Using the WGS approach, Wang et al[10] investigated the microbial community structure and the metabolic variation associated with periodontal health and disease, and found a strong correlation between community structure and disease status. In addition, a number of functional genes and metabolic pathways, including bacterial chemotaxis and glycan biosynthesis, were overrepresented in the microbiomes of periodontal disease. Liu et al[9] and Chen et al[12] investigated bacterial diversity between periodontal health and disease status using 16S rRNA amplicon sequencing and showed that there is a shift in the composition of the oral microbiota between healthy and diseased samples.[13]

In addition to pathogenic microorganisms affecting the underlying supporting structures of teeth, genetic and environmental factors may also contribute to periodontal disease.[14] Prevalence estimates show that the prevalence, severity, and the rate of disease progression vary worldwide.[15] The correlation between the variance in the periodontal microbiome and the prevalence and severity of periodontitis remains unclear.

The aims of this study were to (1) investigate the microbial diversity in subgingival plaques in healthy and diseased people using 16S rRNA metagenomics and quantitative polymerase chain reaction (qPCR), and (2) clarify the variation in the composition of subgingival microbiota in the diseased state in Taiwanese population.

Materials and methods

Ethics statement

This study was approved by the Institutional Review Board of LinKuo Chang Gung Memorial Hospital, Taiwan (Approval no. 102-4239B), and written informed consent was obtained from all the participants involved in this study.

Study participants

Ten healthy individuals and 10 individuals with SP were recruited from the Chang Gung Memorial Hospital. All participants signed an informed consent form prior to their enrolment in the study. Healthy participants with a history of antibiotic treatment in the past 6 months, history of smoking, or diabetes were excluded from the study. The oral health status of all individuals was determined by a dentist who performed a full-mouth clinical examination that included inspection of the teeth, oral mucosa, and periodontal tissues. The healthy participants had normal oral mucous membranes and were free from nonrestored carious lesions. At most sites, periodontal tissues showed no clinical signs of inflammation such as redness, swelling, or breeding on probing and were judged to be free of gingivitis or periodontitis. A diagnosis of SP was determined based on the American Academy of Periodontology parameters.[1] Briefly, SP was defined as more than four sites with CAL $\geq$5 mm. The clinical data of participants with SP are listed in Table S1.

Sample collection and DNA extraction

A total of 20 subgingival plaque samples, including one sample from each of the healthy participants and one sample from each of the SP participants, were obtained. Sampling sites were isolated using cotton rolls after all supragingival plaque, and calculus was removed using sterile Gracey curettes. Teeth were air-dried, and a total of five paper points were inserted into the depth of the pocket for 10 seconds. The paper points were packed into one microcentrifuge tube per tooth, containing 300 μL lysis buffer, and DNA was extracted directly using the MasterPure Gram Positive DNA Purification Kit (Epicenter, Madison, WI, USA). After extraction, DNA was quantified using a fluorometer (Qubit; Invitrogen, Carlsbad, CA, USA). Barcoded PCR amplification was performed using V1 forward primer (5′-AGAGTTTGATCCTGGCTCAG-3′) and V2 reverse primer (5′-TGCTGCCTCCCGTAGGAGT-3′), with 382-bp amplicons flanking the highly variable V1—V2 region of the 16S rRNA gene sequence.[16] PCR reactions were performed using 5× PCR Dye Master Mix II (GeneMark GMbiolab Co., Ltd., Taichung, Taiwan). All PCR cycles included an initial denaturation step at 94°C for 30 seconds, an annealing step at 60°C for 20 seconds, and an extension step at 72°C for 30 seconds. Amplification for 30 cycles was preceded by a one-time denaturation step at 94°C for 3 minutes prior to the first cycle and included a final step at 72°C for 2 minutes to ensure complete extension.

Amplicon sequencing

Individual barcoded PCR products were purified and then pooled at a total combined concentration of 1 μg/50 μL. All barcoded PCR fragments were sequenced using an Illumina Miseq Desktop Sequencer at Yourgene Bioscience Co., Ltd. (Taipei, Taiwan). Raw sequences were deposited in the National Center for Biotechnology Information Sequence Read Archive under the Bioproject accession number PRJNA274944.

Computational and statistical analyses

Paired-end reads were assembled using PEAR software (http://www.exelixis-lab.org/web/software/pear), and low-quality reads were demultiplexed and filtered using split_libraries.py in QIIME with default parameters.[17] The remaining reads were checked for chimeric sequences with UCHIME[18] and clustered into operational taxonomic units (OTUs) using a *de novo* OTU selection protocol at the 97% identity level with the UPARSE algorithm.[19] The taxonomy associated with each OTU was assigned by BLASTing a representative sequence of each OTU against a database combining Greengenes[20] and 16S rRNA datasets obtained from the National Institutes of Health-supported Human Oral Microbiome Database project.[21]

Statistical analyses were performed using R (http://www.R-project.org). The raw number of reads per sample was transformed to relative abundance, and the microbial communities at the phylum and family levels were represented by stacked bar charts. Principal component analysis (PCA) and heat map were performed at the genus level to summarize similarity, and only genera with >1% abundance on average were included. The statistical significance of the observed differences was estimated using Mann—Whitney *U* test.

Quantitative real-time PCR

The amount of *Streptococcus oralis* and *T. forsythia* in a sample was determined by qPCR using an Eco Real-Time PCR System (Illumina, San Diego, CA, USA). Two pairs of primers, FP_SO (5′-AACGGACCCAGACCGTGTTGTAAC-3′) and RP_SO (5′-CCGCCAAGTTGAGGAAGTAGCTCATC-3′) for the *rpoB* gene in *S. oralis* and FP_TF (5′-AACGCAATTCTCGTATGTCCGATAAGGC-3′) and RP_TF (5′-TACCGTTTGTGAGCTCCTGCAACTT-3′) for the *rpoB* gene in *T. forsythia*, were designed using the DesignStudio web-based tool (Illumina). Briefly, the sample DNA extracted was quantitated to 1 ng/μL, and 1 μL of the diluted sample was added to a SYBR Green PCR Master Mix (Biogenesis Technologies, Inc., Taichung, Taiwan) for each reaction. A melting curve analysis was performed to determine the specificity of the PCR products. Genomic DNA from *S. oralis* ATCC35037 and *T. forsythia* ATCC43037 were used to generate standard curves for quantification. All samples were run in triplicate to confirm the reproducibility of the data.

Results

Diversity of bacterial community profiles

Twenty subgingival samples consisting of 10 subgingival samples from healthy participants (designated as ORAL.T1, ORAL.T2, ORAL.T3, ORAL.T4, ORAL.T5, ORAL.T6, ORAL.T7, ORAL.T8, ORAL.T9, and ORAL.T10) and 10 from SP patients (designated as SP001.46, SP002.33, SP003.47, SP004.46, SP005.34, SP006.17, SP007.16, SP008.16, SP009.17, and SP010.24) were collected. The last two numbers in each sample collected from SP denoted the tooth site. After sampling, by DNA extraction and barcoded PCR amplification, a large number of amplicons were obtained and sequenced. In total, the raw dataset of 10 healthy samples and 10 diseased samples contained 404,572 and 1,292,745 sequences, respectively (Table S2). After trimming, we found that 863 and 686 classified sequences phylotypes were in the diseased and the healthy samples, respectively. Additionally, healthy and diseased groups contained 2.4% and 1.3% unclassified phylotypes, respectively. A phylotype is defined here as organisms sharing ≥97% 16S rRNA gene sequence identity.[22]

Figure 1 shows the composition of microbial communities across all subgingival samples. We found a total of six different bacterial phyla: Bacteroidetes, Firmicutes, Fusobacteria, Proteobacteria, Actinobacteria, and Spirochaetes. The difference between the diseased and the healthy samples was derived from the relative proportions of the two abundant phyla, Bacteriodes ($p < 0.01$) and Firmicutes ($p < 0.001$) using Student *t* test. In the healthy samples, Firmicutes (17.8—48.6%) was the most abundant phylum, followed by Bacteroides (12.5—43.3%) and Fusobacteria (7.3—28.6%). In the diseased samples, Bacteroides (36.1—48.8%) was the most abundant phylum followed by Firmicutes (14.9—28.3%) and Fusobacteria (7.7—27.6%)

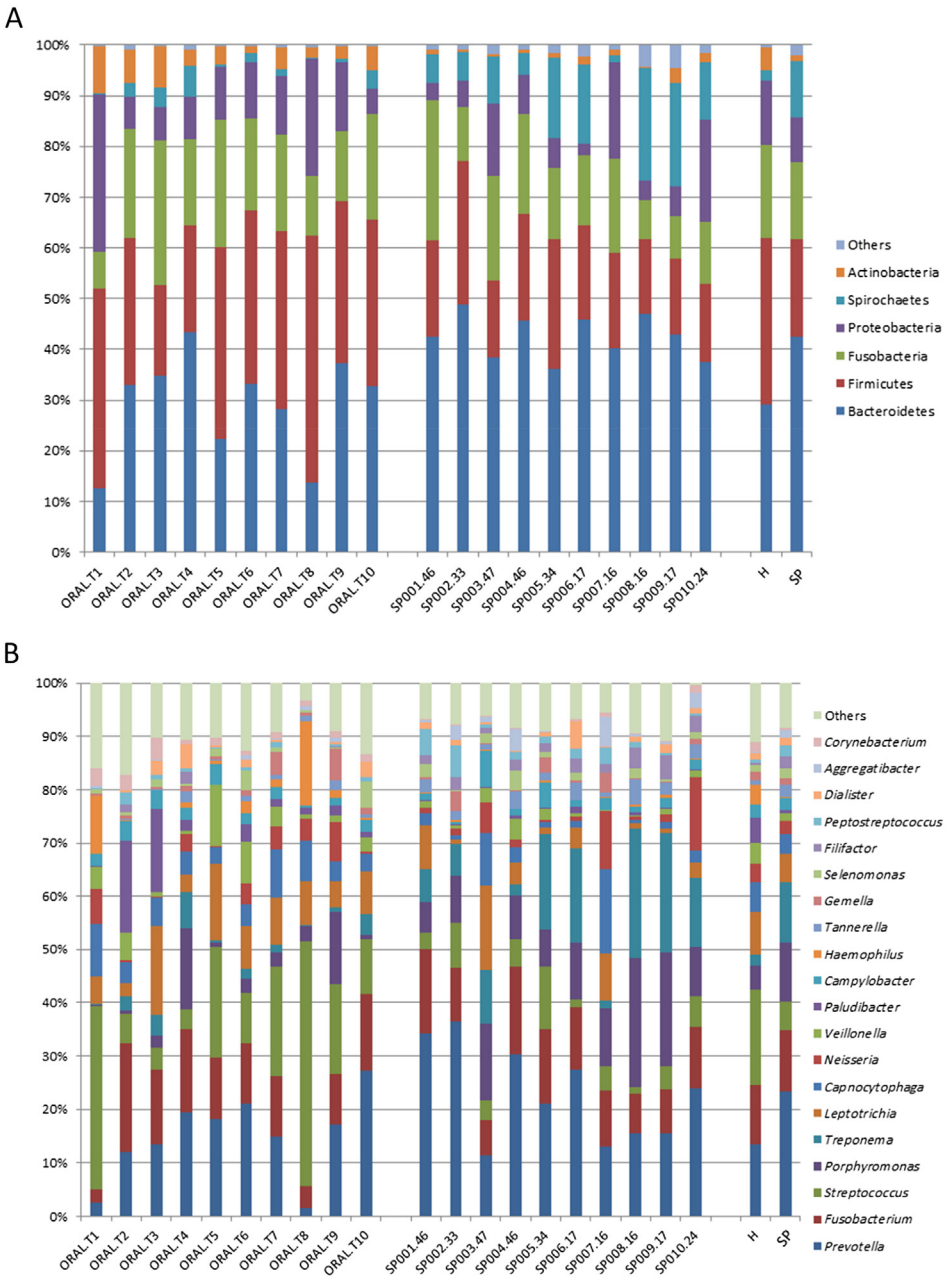


Figure 1. The subgingival bacterial communities of healthy (H) and diseased (SP) individuals. Miseq reads assigned to each (A) phylum and (B) genus are represented with stacked bar charts. The relative height represents the percentage of reads that can be placed at (A) phylum level or (B) genus level using Miseq reads with a BLASTX search of the Greengenes database.

(Figure 1A). At the genus level, differences between the communities in various samples are shown in Figure 1B. A total of 20 different bacterial genera were present across all the samples, with differences in their distribution. *Streptococcus* were prevalent across most of the healthy samples compared with the diseased samples. In contrast, *Prevotella*, *Porphyromonas*, and *Treponema* were abundant in most of the diseased samples.

Connections between bacterial community profiles

To further explore the relationships between different bacterial communities in all samples, a PCA analysis was performed using genus-level taxonomic profiles. As shown in Figure 2A, the first two principal components, representing 47.6% of the variance, classified the 20 samples into two groups, with nine healthy samples forming a group that was separated from the diseased group. P1 comprised 11 genus types, including *Filibacter*, *Tannerella*, *Treponema*, *Porphyromonas*, *Prevotella*, *Veilonella*, *Haemophilus*, *Leptotrichia*, *Capnocytophaga*, *Corynebacterium*, and *Streptococcus*. P2 consisted of five genus types, including *Fusobacterium*, *Paludibacter*, *Selenomas*, *Haemophilus*, and *Neisseria*. Interestingly, three samples—one from a healthy sample (ORAL.T4) and two diseased samples (SP003.47 and SP007.16)—were separated from each

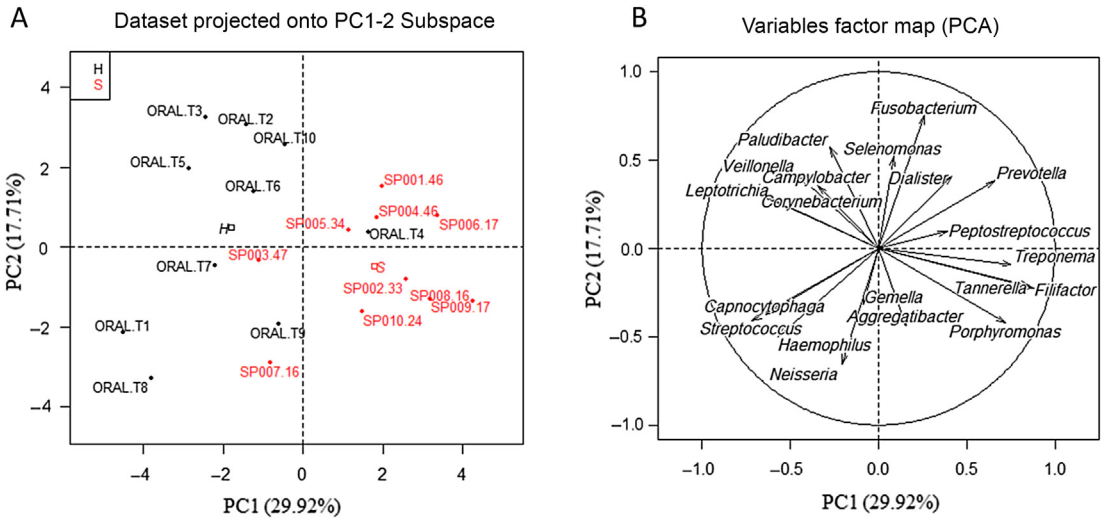


Figure 2. Clustering relationships of bacterial community in periodontal health and disease. (A) Principal component analysis (PCA) of bacterial communities at the genera level from healthy (black) and diseased (red) individuals. (B) Variable factor map at the genera level.

group. Cluster analysis by heat map revealed that sample ORAL T4 was clustered into the diseased samples, as shown in Figure 3. A closer investigation of the bacterial community in sample ORAL T4 showed that *Porphyromonas* was the most abundant and *Streptococcus* was the least abundant across all healthy samples, despite the clinical parameters for evaluation of periodontitis being quite normal. Additionally, Sample SP007.16 and SP003.47 were clustered into the healthy group. *Capnocytophaga*, *Neisseria*, and *Leptotrichia* were more abundant in the two samples compared with other diseased samples. For a closer investigation of the bacterial differences across all the samples, a variable factor map was produced using R software, and the result is shown in Figure 2B.

Figure 4 shows subgingival community among healthy and diseased groups. A total of 13 genera were found in both groups of samples. Notably, two periodontal genera previously classified as "red complex"[4] including *Porphyromonas* and *Treponema* were found in all samples. Moreover, except for *Peptostreptococcus*, four genera previously classified as "orange complex"[4] (*Campylobacter*, *Fusobacterium*, *Prevotella*, and *Streptococcus*) were also

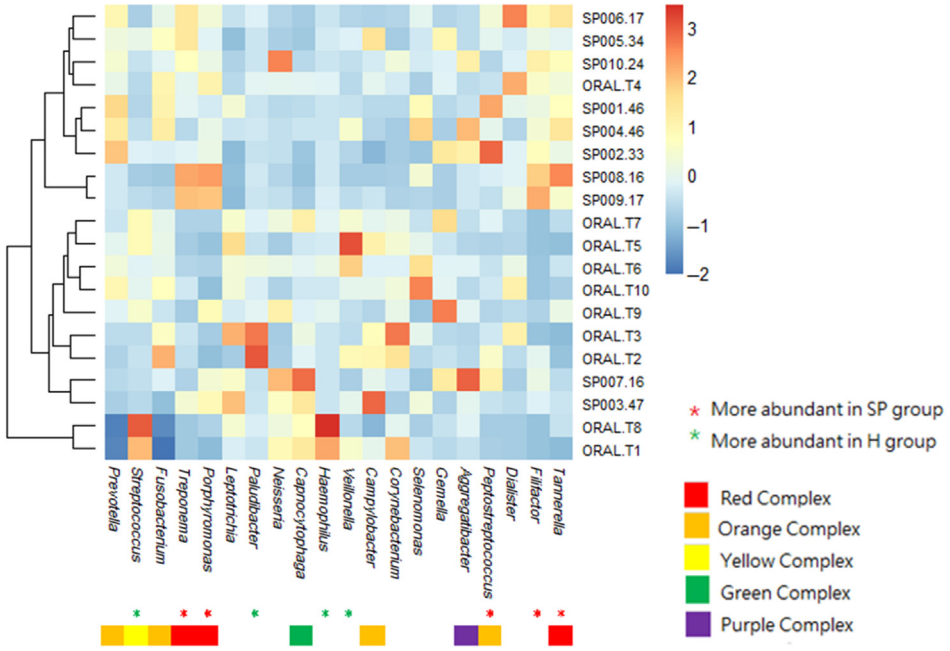


Figure 3. Relative abundance of genera in the sample estimated from 16S rRNA amplicon sequencing. Only genera with >1% abundance on average were included. Color reflects relative abundance from low (blue) to high (red). Organisms are colored according to the bacterial complex designations of Socransky et al.[4] H = healthy; SP = severe chronic periodontitis.

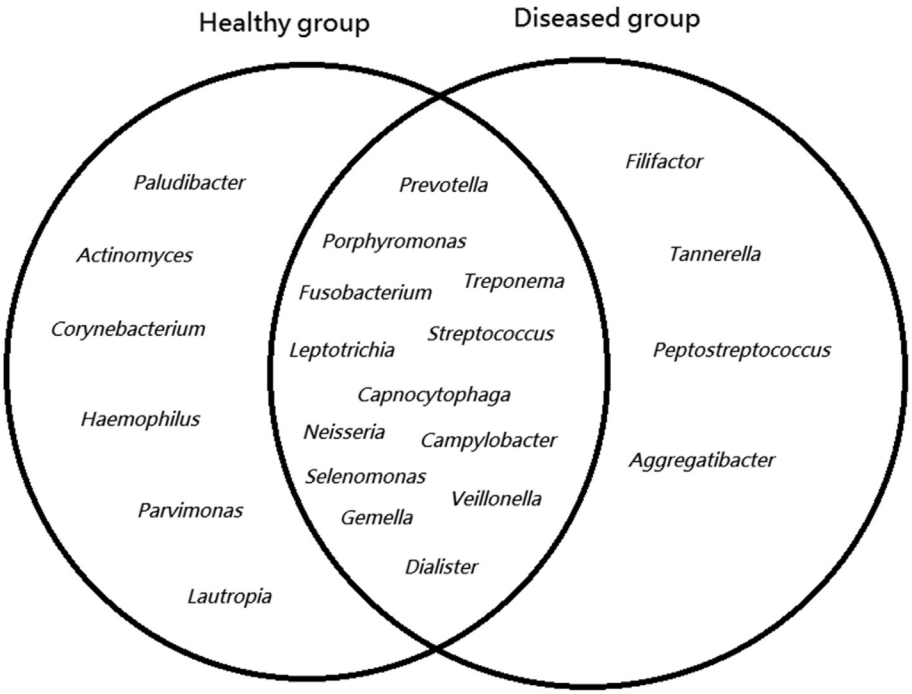


Figure 4. Subgingival community among healthy and diseased groups. Only genera with >1% abundance on average were included.

present in all 20 samples. Four genera—*Filifactor*, *Tannerella*, *Peptostreptococcus*, and *Aggregatibacter*—were only present in the diseased group, whereas six genera—*Paludibacter*, *Actinomyces*, *Corynebacterium*, *Haemophilus*, *Parvimonas*, and *Lautropia*—were only observed in the healthy group.

To clarify the difference in bacterial composition between the healthy and diseased samples, a total of 14 genera, including eight genera that dominated in the healthy samples and six genera enriched in the diseased samples, were selected, with $p \leq 0.05$ (Table 1). Notably, *Lautropia* ($p = 0.0001$) and *Filifactor* ($p = 0.001$) were more significantly different in the healthy samples and the diseased samples, respectively.

Evaluation of the abundance of bacterial species by qPCR

Although 16S rRNA metagenomics is important for identifying a strain to the family or genus level, it is of little utility for differentiation of species.[23] Many bacteria harbor multiple, heterogeneous rRNA operons,[24] and it can lead to an overestimation of microbial diversity with 16S rRNA approaches.[25] To clarify the abundance of the bacteria in a sample, the quantitative measurement of *rpoB* using qPCR was an alternative approach as each bacterium contains one copy of *rpoB*.In addition, qPCR is a rapid and sensitive approach for species identification. Figure 5 shows the distribution of *S. oralis* and *T. forsythia* across all subgingival samples. The trend of abundance for these two microorganisms using qPCR in all samples was consistent with the result obtained by 16S rRNA metagenomics (Figure 5).

Table 1 Significantly different genera between healthy and diseased participants.

Genus[a]	H (%) ($n = 10$)	SP (%) ($n = 10$)	p[b]
Healthy group			
Lautropia	1.00	0.00	0.0001
Corynebacterium	1.82	0.32	0.000
Parvimonas	1.25	0.08	0.0069
Streptococcus	17.16	4.90	0.0115
Actinomyces	1.19	0.39	0.0126
Haemophilus	3.43	0.26	0.0140
Capnocytophaga	5.41	3.75	0.0232
Paludibacter	4.43	0.43	0.0232
Diseased group			
Filifactor	0.42	2.47	0.0010
Tannerella	0.75	2.38	0.0015
Treponema	2.26	12.00	0.0029
Porphyromonas	4.18	12.04	0.0068
Aggregatibacter	0.34	1.81	0.0311
Peptostreptococcus	0.66	2.08	0.0342

[a] Only genera with >1% abundance were included.
[b] Statistical significance was estimated using Mann—Whitney *U* test.
H = healthy; SP = severe chronic periodontitis.

Discussion

Periodontitis is a bacterially induced chronic inflammatory disease of the periodontium. With the advent of high-throughput DNA sequencing technology, which offers a resolution several orders of magnitude higher than that

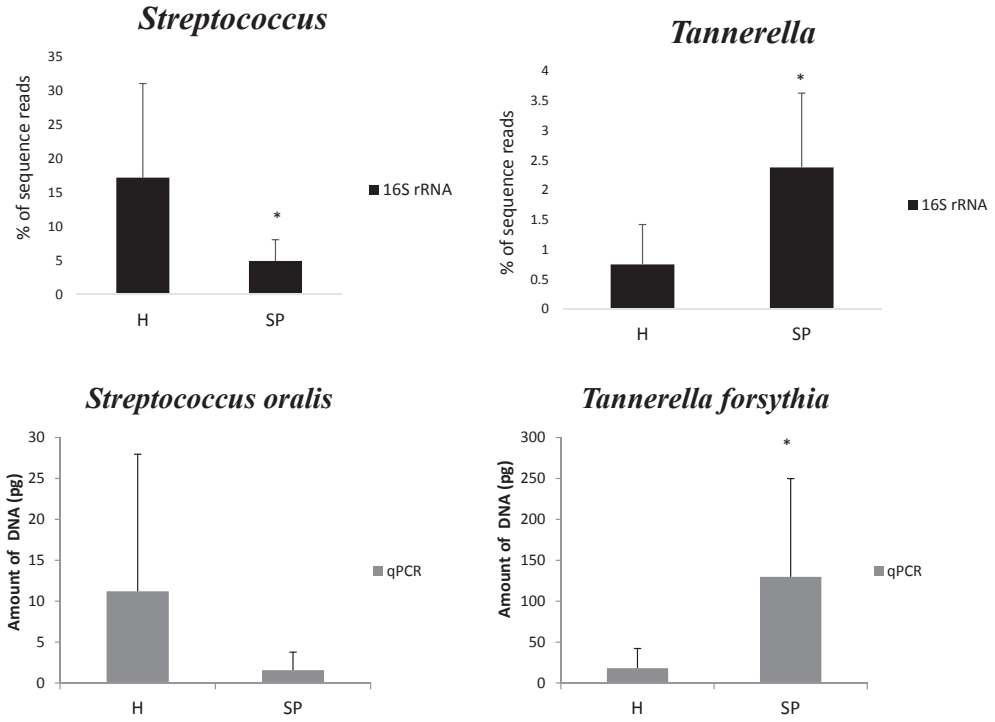


Figure 5. Distribution between *Streptococcus oralis* (SO) and *Tannerella forsythia* (TF) in healthy (H) and diseased (SP) samples. Bacterial distribution among the two groups analyzed by 16S rRNA metagenomics and qPCR. qPCR = quantitative polymerase chain reaction. *$p < 0.05$.

offered by cultivable approaches, research into bacterial community composition is greatly facilitated.[26] This case—control study, carried out using Illumina Miseq Desktop Sequencing of subgingival samples from 10 individuals with SP and 10 periodontally health controls, showed differences in community composition. This study confirms previous findings that a microbial shift occurs during the transition from periodontal health to periodontal disease[13] and provides a much clearer picture of the overall community complexity in periodontal disease.

In the present study, we report several findings. First, we observed a higher diversity of bacteria in diseased samples (863 phylotypes) compared with healthy samples (686 phylotypes). Second, the major difference between the diseased and the healthy samples was derived from the relative proportion of three abundant phyla—Bacteroidetes, Firmicutes, and Fusobacteria. Bacteroidetes was the most abundant phylum in all diseased samples, whereas its abundance decreased in healthy samples. Instead, Firmicutes was the most abundant phylum in healthy samples. Third, despite 13 genera being found in all samples, the abundance of three genera—*Porphyromonas* and *Treponema* (recognized as the "red complex") and *Streptococcus* (recognized as the "orange complex" described by Socransky et al[4])—was significantly different between diseased and healthy samples ($p < 0.05$). Fourth, the healthy- and diseased-associated communities are highly different in PCA and heat map analyses.

Currently, several reports have documented the bacterial community in patients with chronic periodontitis and healthy control participants using metagenomics.[9,10,27] The core microbiota associated with chronic periodontitis and periodontally healthy participants are largely consistent in those studies; however, we observed several discrepancies. First, the relative abundance of the most dominant genera differed among previous studies. For example, Griffen et al[27] reported that *Prevotella* and *Fusobacterium* were the most abundant genera in participants, using different primers and target regions (V1—V2 or V4 of the 16S rRNA gene, respectively). Liu et al[9] showed that neither *Fusobacterium* nor *Porphyromonas* were abundant in periodontal disease samples using WGS sequencing, in spite of being previously implicated in this disease.[13] They also characterized the genomes of key players in the subgingival microbiota in periodontitis patients, including an unculturable TM7 organism. Abusleme et al[28] found *Prevotella* to have lower abundance, unchanged between groups. By contrast, our study showed that *Porphyromonas* demonstrated significant difference between groups, as shown in Table 1. In addition, we did not detect TM7 in diseased participants. This discrepancy could be explained by geographic variability[29] or by difference in the depths of the pockets sampled,[4] as well as the sample size and the DNA analytic bias.[30] Second, a large increase in the proportion of Gram-negative bacteria in diseased samples compared with healthy samples may account for the decreases in the proportion of the health-associated genus *Streptococcus*.[10,27] However, at least two Gram-positive bacteria, *Filifactor* and *Peptostreptococcus*, may be correlated with severe disease in our study and those of other groups,[11,27,31] but not in WGS studies.[9,10] One possible explanation is that metagenomic shotgun sequencing may generate 16S rRNAs more randomly, which is more robust to differences in community structure.[32]

In this study, we performed qPCR to identify species and quantitate the amount of DNA within a sample via the *rpoB* gene. *S. oralis* and *T. forsythia* were chosen as species candidates based on two reports showing that *S. oralis* was the most abundant *Streptococcus* species after periodontal therapy,[11] whereas *T. forsythia* was one of the major periodontal pathogens in periodontitis participants.[11,33] Our results indicated similar trends of abundance using both methods. However, a closer investigation of the two species analyzed by qPCR and 16S rRNA metagenomics reveals quite different results. Compared with *T. forsythia*, the high abundance of *Streptococcus* spp. in healthy or diseased samples corresponded to low amounts of *S. oralis* DNA, indicating that *Streptococcus* spp. may include several species in addition to *S. oralis*. A previous report found that three *Streptococcus* species—*Streptococcus mitis*, *Streptococcus sanguinis*, and *Streptococcus intermedius*—were enriched in healthy samples,[27] implying that those organisms may contribute to the change of relative proportions in SP. The microbial shift from healthy to diseased state in species levels will necessitate further investigation using qPCR. Nevertheless, our results suggested that the species identification and the abundance estimates within samples by qPCR could compensate for the disadvantages of 16S rRNA metagenomics.

In conclusion, this study provides a picture of the structure of the microbiome in patients with SP and healthy Taiwanese people. We demonstrate a microbial shift from health to disease that could provide the basis for further understanding host—microbe interactions in periodontitis.

Conflicts of interest

All contributing authors declare no conflicts of interest.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jmii.2016.04.007.